

Aim : Create a simple Spark MLlib pipeline to classify data. Use logistic regression or decision trees.

CODE :

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import when
from pyspark.ml import Pipeline
from pyspark.ml.feature import VectorAssembler
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
```

```
spark = SparkSession.builder \
    .appName("FIFA Classification") \
    .master("local[*]") \
    .getOrCreate()
```

```
data = spark.read \
    .option("header", "true") \
    .option("inferSchema", "true") \
    .csv("fifa.csv")
```

```
print("FIFA Data:")
data.show()
```

```
data = data.withColumn(
    "label",
    when(data["ARG"] <= 10, 1.0).otherwise(0.0)
)
```

```
print("Data with Label:")
data.show()
```

```
assembler = VectorAssembler(
    inputCols=["BRA", "ESP", "FRA", "GER", "ITA"],
    outputCol="features"
)
```

```
logistic_regression = LogisticRegression(
    labelCol="label",
    featuresCol="features",
    maxIter=10
)
```

```
pipeline = Pipeline(
    stages=[assembler, logistic_regression]
)
```

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```
training_data, test_data = data.randomSplit(  
    [0.8, 0.2],  
    seed=42  
)
```

```
print("Training Data:")  
training_data.show()
```

```
print("Test Data:")  
test_data.show()
```

```
model = pipeline.fit(training_data)
```

```
predictions = model.transform(test_data)
```

```
print("Predictions:")  
predictions.select(  
    "ARG",  
    "BRA",  
    "ESP",  
    "FRA",  
    "GER",  
    "ITA",  
    "label",  
    "prediction"  
)show()
```

```
evaluator = MulticlassClassificationEvaluator(  
    labelCol="label",  
    predictionCol="prediction",  
    metricName="accuracy"  
)
```

```
accuracy = evaluator.evaluate(predictions)
```

```
print("Classification Accuracy =", accuracy)
```

```
spark.stop()
```

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OUTPUT :

```
Run omwala_M2_pract13 x
```

```
26/09/21 12:26:45 WARN NativeCodeLoader: Unable to load native code
FIFA Data:
+-----+-----+-----+-----+-----+-----+
|      Date| ARG|BRA| ESP| FRA|GER| ITA|
+-----+-----+-----+-----+-----+-----+
|1993-08-08| 5.0|8.0|13.0|12.0|1.0| 2.0|
|1993-09-23|12.0|1.0|14.0| 7.0|5.0| 2.0|
|1993-10-22| 9.0|1.0| 7.0|14.0|4.0| 3.0|
|1993-11-19| 9.0|4.0| 7.0|15.0|3.0| 1.0|
|1993-12-23| 8.0|3.0| 5.0|15.0|1.0| 2.0|
|1994-02-15| 9.0|2.0| 6.0|14.0|1.0| 7.0|
|1994-03-15| 8.0|2.0| 6.0|15.0|1.0|11.0|
|1994-04-19|10.0|1.0| 7.0|15.0|2.0|13.0|
|1994-05-17| 6.0|1.0| 9.0|17.0|2.0|16.0|
|1994-06-14| 8.0|3.0| 5.0|13.0|1.0| 4.0|
|1994-07-21| 9.0|1.0| 6.0|16.0|4.0| 2.0|
|1994-09-13| 9.0|1.0| 6.0|20.0|4.0| 2.0|
|1994-10-25| 9.0|1.0| 6.0|19.0|4.0| 2.0|
|1994-11-22| 9.0|1.0| 5.0|18.0|4.0| 3.0|
|1994-12-20|10.0|1.0| 2.0|19.0|5.0| 4.0|
|1995-02-20| 7.0|1.0| 2.0|18.0|5.0| 3.0|
|1995-04-19| 8.0|1.0| 3.0|18.0|4.0| 2.0|
|1995-05-16| 8.0|1.0| 3.0|17.0|5.0| 2.0|
|1995-06-13|10.0|1.0| 3.0|17.0|5.0| 2.0|
|1995-07-25| 5.0|1.0| 3.0|18.0|4.0| 6.0|
+-----+-----+-----+-----+-----+-----+
only showing top 20 rows
```

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Data with Label:

Date	ARG	BRA	ESP	FRA	GER	ITA	label
1993-08-08	5.0	8.0	13.0	12.0	1.0	2.0	1.0
1993-09-23	12.0	1.0	14.0	7.0	5.0	2.0	0.0
1993-10-22	9.0	1.0	7.0	14.0	4.0	3.0	1.0
1993-11-19	9.0	4.0	7.0	15.0	3.0	1.0	1.0
1993-12-23	8.0	3.0	5.0	15.0	1.0	2.0	1.0
1994-02-15	9.0	2.0	6.0	14.0	1.0	7.0	1.0
1994-03-15	8.0	2.0	6.0	15.0	1.0	11.0	1.0
1994-04-19	10.0	1.0	7.0	15.0	2.0	13.0	1.0
1994-05-17	6.0	1.0	9.0	17.0	2.0	16.0	1.0
1994-06-14	8.0	3.0	5.0	13.0	1.0	4.0	1.0
1994-07-21	9.0	1.0	6.0	16.0	4.0	2.0	1.0
1994-09-13	9.0	1.0	6.0	20.0	4.0	2.0	1.0
1994-10-25	9.0	1.0	6.0	19.0	4.0	2.0	1.0
1994-11-22	9.0	1.0	5.0	18.0	4.0	3.0	1.0
1994-12-20	10.0	1.0	2.0	19.0	5.0	4.0	1.0
1995-02-20	7.0	1.0	2.0	18.0	5.0	3.0	1.0
1995-04-19	8.0	1.0	3.0	18.0	4.0	2.0	1.0
1995-05-16	8.0	1.0	3.0	17.0	5.0	2.0	1.0
1995-06-13	10.0	1.0	3.0	17.0	5.0	2.0	1.0
1995-07-25	5.0	1.0	3.0	18.0	4.0	6.0	1.0

only showing top 20 rows

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Training Data:

```
+-----+-----+-----+-----+-----+-----+
|      Date| ARG|BRA| ESP| FRA|GER| ITA|label|
+-----+-----+-----+-----+-----+-----+
|1993-08-08| 5.0|8.0|13.0|12.0|1.0| 2.0| 1.0|
|1993-09-23|12.0|1.0|14.0| 7.0|5.0| 2.0| 0.0|
|1993-11-19| 9.0|4.0| 7.0|15.0|3.0| 1.0| 1.0|
|1993-12-23| 8.0|3.0| 5.0|15.0|1.0| 2.0| 1.0|
|1994-02-15| 9.0|2.0| 6.0|14.0|1.0| 7.0| 1.0|
|1994-04-19|10.0|1.0| 7.0|15.0|2.0|13.0| 1.0|
|1994-06-14| 8.0|3.0| 5.0|13.0|1.0| 4.0| 1.0|
|1994-07-21| 9.0|1.0| 6.0|16.0|4.0| 2.0| 1.0|
|1994-09-13| 9.0|1.0| 6.0|20.0|4.0| 2.0| 1.0|
|1994-10-25| 9.0|1.0| 6.0|19.0|4.0| 2.0| 1.0|
|1994-12-20|10.0|1.0| 2.0|19.0|5.0| 4.0| 1.0|
|1995-02-20| 7.0|1.0| 2.0|18.0|5.0| 3.0| 1.0|
|1995-04-19| 8.0|1.0| 3.0|18.0|4.0| 2.0| 1.0|
|1995-05-16| 8.0|1.0| 3.0|17.0|5.0| 2.0| 1.0|
|1995-06-13|10.0|1.0| 3.0|17.0|5.0| 2.0| 1.0|
|1995-08-22| 5.0|1.0| 3.0|18.0|4.0| 6.0| 1.0|
|1995-09-19| 9.0|1.0| 6.0|20.0|4.0| 2.0| 1.0|
|1995-10-17| 7.0|1.0| 2.0|12.0|3.0| 6.0| 1.0|
|1995-12-19| 7.0|1.0| 4.0| 8.0|2.0| 3.0| 1.0|
|1996-01-24| 7.0|1.0| 4.0|13.0|2.0| 3.0| 1.0|
+-----+-----+-----+-----+-----+-----+
only showing top 20 rows
```

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Test Data:

```
+-----+-----+-----+-----+-----+-----+-----+
|      Date| ARG|BRA| ESP| FRA| GER| ITA|label|
+-----+-----+-----+-----+-----+-----+-----+
|1993-10-22| 9.0|1.0| 7.0|14.0| 4.0| 3.0| 1.0|
|1994-03-15| 8.0|2.0| 6.0|15.0| 1.0|11.0| 1.0|
|1994-05-17| 6.0|1.0| 9.0|17.0| 2.0|16.0| 1.0|
|1994-11-22| 9.0|1.0| 5.0|18.0| 4.0| 3.0| 1.0|
|1995-07-25| 5.0|1.0| 3.0|18.0| 4.0| 6.0| 1.0|
|1995-11-21| 7.0|1.0| 3.0| 9.0| 2.0| 4.0| 1.0|
|1996-07-03|15.0|1.0| 8.0| 3.0| 2.0| 5.0| 0.0|
|1997-02-27|18.0|1.0| 7.0| 3.0| 2.0|10.0| 0.0|
|1998-02-18|17.0|1.0|23.0| 6.0| 2.0|14.0| 0.0|
|1998-03-18| 7.0|1.0|25.0|14.0| 2.0|13.0| 1.0|
|1998-04-22| 7.0|1.0| 9.0|25.0| 2.0|16.0| 1.0|
|1998-07-15| 5.0|1.0|14.0| 2.0| 3.0| 8.0| 1.0|
|1998-09-16| 5.0|1.0|14.0| 2.0| 3.0| 7.0| 1.0|
|1999-01-27| 6.0|1.0| 9.0| 2.0| 5.0| 4.0| 1.0|
|1999-08-18| 6.0|1.0| 7.0| 2.0| 5.0| 4.0| 1.0|
|2000-03-15| 7.0|1.0| 4.0| 3.0| 5.0|14.0| 1.0|
|2000-11-15| 3.0|1.0| 6.0| 2.0| 8.0| 5.0| 1.0|
|2002-11-20| 5.0|1.0| 3.0| 2.0| 4.0|11.0| 1.0|
|2003-11-19| 4.0|1.0| 3.0| 2.0| 9.0|11.0| 1.0|
|2004-03-17| 6.0|1.0| 3.0| 2.0|10.0|11.0| 1.0|
+-----+-----+-----+-----+-----+-----+-----+
only showing top 20 rows
```

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Predictions:

```
+---+---+---+---+---+---+---+---+
| ARG|BRA| ESP| FRA| GER| ITA|label|prediction|
+---+---+---+---+---+---+---+---+
| 9.0|1.0| 7.0|14.0| 4.0| 3.0| 1.0| 1.0|
| 8.0|2.0| 6.0|15.0| 1.0|11.0| 1.0| 1.0|
| 6.0|1.0| 9.0|17.0| 2.0|16.0| 1.0| 1.0|
| 9.0|1.0| 5.0|18.0| 4.0| 3.0| 1.0| 1.0|
| 5.0|1.0| 3.0|18.0| 4.0| 6.0| 1.0| 1.0|
| 7.0|1.0| 3.0| 9.0| 2.0| 4.0| 1.0| 1.0|
|15.0|1.0| 8.0| 3.0| 2.0| 5.0| 0.0| 1.0|
|18.0|1.0| 7.0| 3.0| 2.0|10.0| 0.0| 1.0|
|17.0|1.0|23.0| 6.0| 2.0|14.0| 0.0| 1.0|
| 7.0|1.0|25.0|14.0| 2.0|13.0| 1.0| 1.0|
| 7.0|1.0| 9.0|25.0| 2.0|16.0| 1.0| 1.0|
| 5.0|1.0|14.0| 2.0| 3.0| 8.0| 1.0| 1.0|
| 5.0|1.0|14.0| 2.0| 3.0| 7.0| 1.0| 1.0|
| 6.0|1.0| 9.0| 2.0| 5.0| 4.0| 1.0| 1.0|
| 6.0|1.0| 7.0| 2.0| 5.0| 4.0| 1.0| 1.0|
| 7.0|1.0| 4.0| 3.0| 5.0|14.0| 1.0| 1.0|
| 3.0|1.0| 6.0| 2.0| 8.0| 5.0| 1.0| 1.0|
| 5.0|1.0| 3.0| 2.0| 4.0|11.0| 1.0| 1.0|
| 4.0|1.0| 3.0| 2.0| 9.0|11.0| 1.0| 1.0|
| 6.0|1.0| 3.0| 2.0|10.0|11.0| 1.0| 1.0|
+---+---+---+---+---+---+---+---+
```

only showing top 20 rows

Classification Accuracy = 0.9285714285714286